DO NOW — RECALL

Lesson 083 · Dijkstra: Shortest Paths with Non-negative

Name: _____ Class: _____ Date: _____

1. What does relaxation do?

2. What is a weighted edge?

3. What condition does standard Dijkstra require?

4. Undirected graph edges A–B, A–C, B–D. Give BFS order from A visiting B before C.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Updates a tentative distance via a cheaper path.

2. Model answer

Connection with a cost.

3. Model answer

Non-negative edge weights.

4. Model answer

A,B,C,D.

By the end, you can…

• Dijkstra maintains tentative shortest distances from a start vertex. All edge weights must be non-negative.

• Repeatedly finalise the unsettled node with smallest tentative distance, then relax outgoing edges to improve neighbour estimates.

• Track predecessor nodes alongside distances to reconstruct the shortest route, not just its total cost.

Vocabulary — know these terms

Dijkstra — Non-negative weighted shortest paths.

tentative distance — Current best-known route cost.

relaxation — Replace distance if improved.

predecessor — Previous vertex on best path.

NEW LEARNING

Read, annotate and explain

Observe

Dijkstra maintains tentative shortest distances from a start vertex. All edge weights must be non-negative.

Apply

Repeatedly finalise the unsettled node with smallest tentative distance, then relax outgoing edges to improve neighbour estimates.

Evaluate

Track predecessor nodes alongside distances to reconstruct the shortest route, not just its total cost.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Find the shortest P→R route. Directed edges: P→Q = 3, P→R = 8, Q→R = 2.

Update neighbours

Start with dist(P)=0 . Settle P (smallest unsettled value). Relax P→Q: Q=3 , predecessor Q←P. Relax P→R: R=8 , predecessor R←P. Remaining unsettled distances: Q=3, R=8.

Improve the tentative route

Q=3 is the smallest unsettled distance. Via Q→R, the candidate is 3+2=5 . This beats R=8, so update R=5 and predecessor R←Q. Settled now: P,Q.

Reconstruct the actual path

R=5 is next, and is the target. Follow predecessor pointers R←Q←P ; reverse to obtain P→Q→R , total cost 5 . No vertex with a larger tentative distance was prematurely finalised.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

From A in stated graph, give tentative distances after settling A and B. [6]

Question 2

Continue from C: give final shortest cost and predecessor chain for D. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why negative edge weights can invalidate Dijkstra's finalisation rule, using a counterexample. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Dijkstra is guaranteed correct on graphs with arbitrary negative edge costs.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.